

1 Basics of .NET development

The .NET framework was introduced by Microsoft in June 2000 with a vision for embracing the internet. At the heart of its strategy was language and platform independence which can be arguably denied because of its close tie-up with the Windows operating system. But what it did give you was the ability to use various languages like Visual Basic .NET, Visual C++ .NET, C#, IronPython and more, and executes your program irrespective of the language you used with the Common Language Runtime (CLR). .NET programs compiled to Microsoft Intermediate Language (MSIL), which is then translated into machine code and executed. There is a Linux port developed by the Mono project, but .NET is still primarily used for Windows development. As the saying goes “If you are not programming for Windows, .NET is not your cup of tea”.

1.1 C#

It is an event driven, object oriented, visual programming language with roots in C, C++ and Java which was developed at Microsoft by Anders Hejlsberg *et al.* before being integrated into the .NET platform. What this gives is the ability to access your programs through any device which supports the CLR, distribute web-based applications through the ASP. Net platform and communicate with other .NET compatible computer languages. Another advantage of C# over other contemporary languages is the strong integration into one of the best Integrated Design Environments (IDE) ever made, i.e., Visual Studio. Although the role of an IDE in language preference can be debated, you cannot disagree with the fact that Visual Studio makes programming and debugging fast and easy and helps you with Rapid Application Development (RAD).

Some of the advantages of using C# over other languages are:

Its supports so-called safe internet programming or managed code which restricts your programs by sand-boxing them but makes sure that your computer/OS is not affected adversely by poorly written code - no core dump or dead console.

- It is simpler than other object-oriented languages (C++).
- It has a very good graphics package and an excellent designer in the Form of Visual Studio and Expression Studio.
- It is related to C and C++, and therefore easy to pick up and start